Quick-start guide to experimental design

PSYC 11: Laboratory in Psychological Science

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What is the purpose of running an experiment?

Goals of experimentation

- Understand or explore how something works
- Distinguish between several potential alternatives
- Get more information (data!)

What "counts" as an experiment?

Types of experiments

- Observing something happening without manipulating it
- Self-reflection or self-report
- Constructing a scenario in which the process you want to study occurs
- Running a simulation (via real-life or computation)

Observational studies

Observe without interfering

- **Prime directive:** do not interfere
- Go out into the world and collect data
- Examples:
  - Studying an animal colony
  - Watch how people interact at a party
  - Record facial expressions as people enter and exit a movie screening
  - Measure highway traffic density at different times of day

Self-report

Trust people to report their experiences

- Socrates: "To know thyself is the beginning of wisdom"
- Trust people to accurately report their opinions, thoughts, feelings, and behaviors
- Examples:
  - Voter preference/opinion surveys
  - Mental health screening forms
  - Feedback surveys
  - Personal narratives

Constructed scenarios

Two main approaches

- In general, there are two widely used approaches:
  - **Classic:** maximize **control**
  - **Naturalistic:** maximize **realism**

Classic experiments

Maximizing control

- Figure out the simplest possible version of the process or phenomenon you're interested in
- If you think the process depends on participants' experiences (e.g., stimuli, framing, effort, etc.), carefully control those factors across several conditions
- Use behaviors (across conditions if needed) as a stand-in for the "real thing"

Classic experiments: examples

Simplified paradigms

- **Vision:** can you see this tiny spot of light?
- **Audition:** what word do you hear when I play this sound?
- **Memory:** which images in this set do you recognize?
- **Cognitive control:** how tightly can you regulate your responses to a cue?

Naturalistic experiments

Maximizing realism

- Create a scenario that closely approximates the real-world process you want to study
- Measure as much stuff as possible about people's behaviors
- Mine the data for patterns

Naturalistic experiments: examples

Rich, real-world paradigms

- **Vision:** where do you fixate at each moment while moving around in a VR environment?
- **Memory:** how you describe "what happened" in a movie?
- **Motor control:** how do people adjust their gait on different terrains?

Simulations

Two types of simulations

- **Real-world:** use props, actors, and technology to make it seem like the participant is in a particular situation, or that the world works a particular way
- **Simulated:** change the properties of a simulated environment and study an artificial agent's behaviors

Simulations: examples

Simulation paradigms

- Confederate experiments
- Misleading participants about the situation or rules
- Simulating how robots of different configurations complete obstacle courses under different gravitational loads

Implementation tools

Tools for building experiments

- Notebooks, audiovisual recordings
- Google forms, qualtrics
- Slideshows
- PsychoPy, jsPsych
- Google Colaboratory

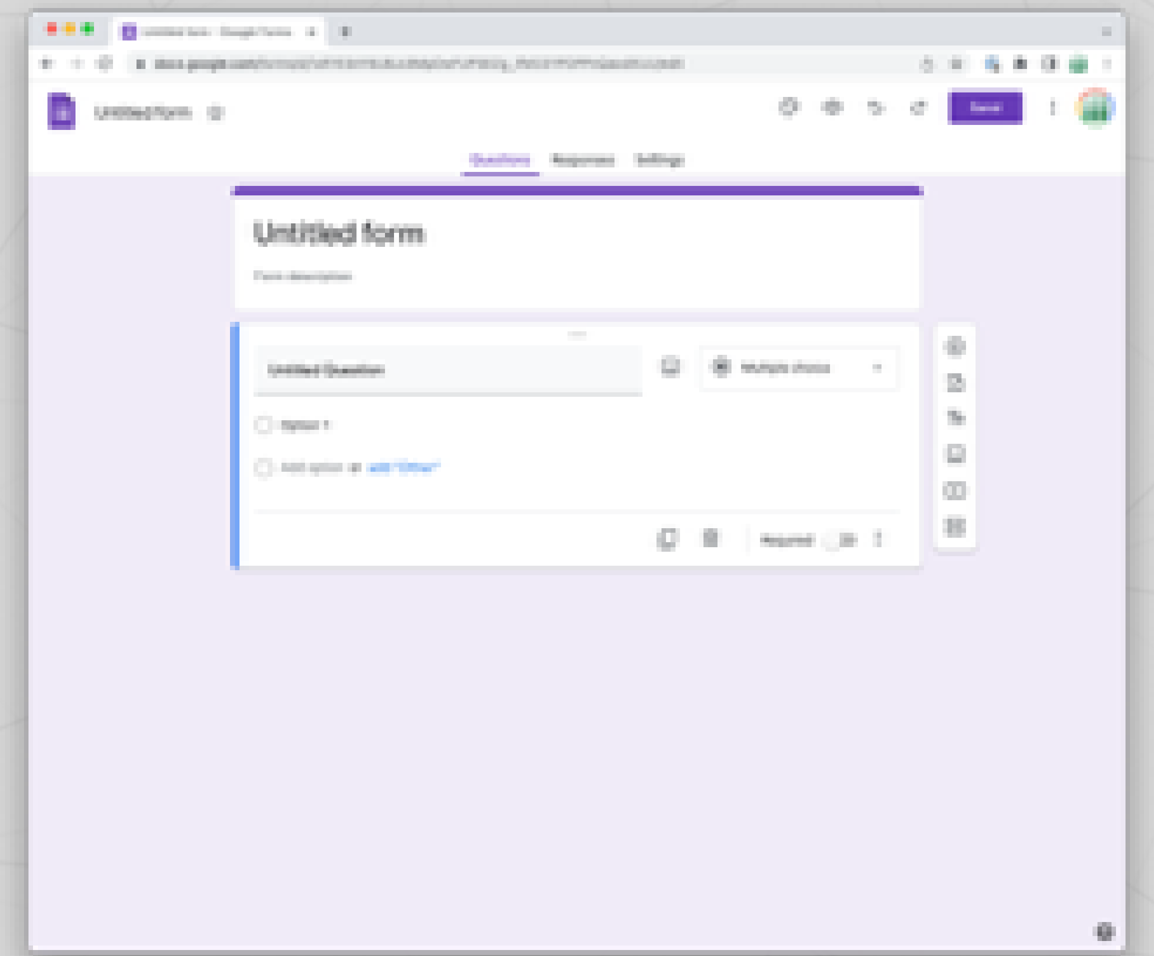
Notebooks and recordings

Low-tech data collection

- Good for observational studies
- Collect lots of richly structured data!
- No setup required!
- Can be tough to analyze (either time-consuming to annotate, or tricky to design effective automated approaches)

Google forms and qualtrics

Slideshows



PsychoPy

jsPsych

Other considerations

Practical advice

- Use what you already know
- Simplicity: the art of maximizing the amount of work **not** done
- Work together
- Ask for help